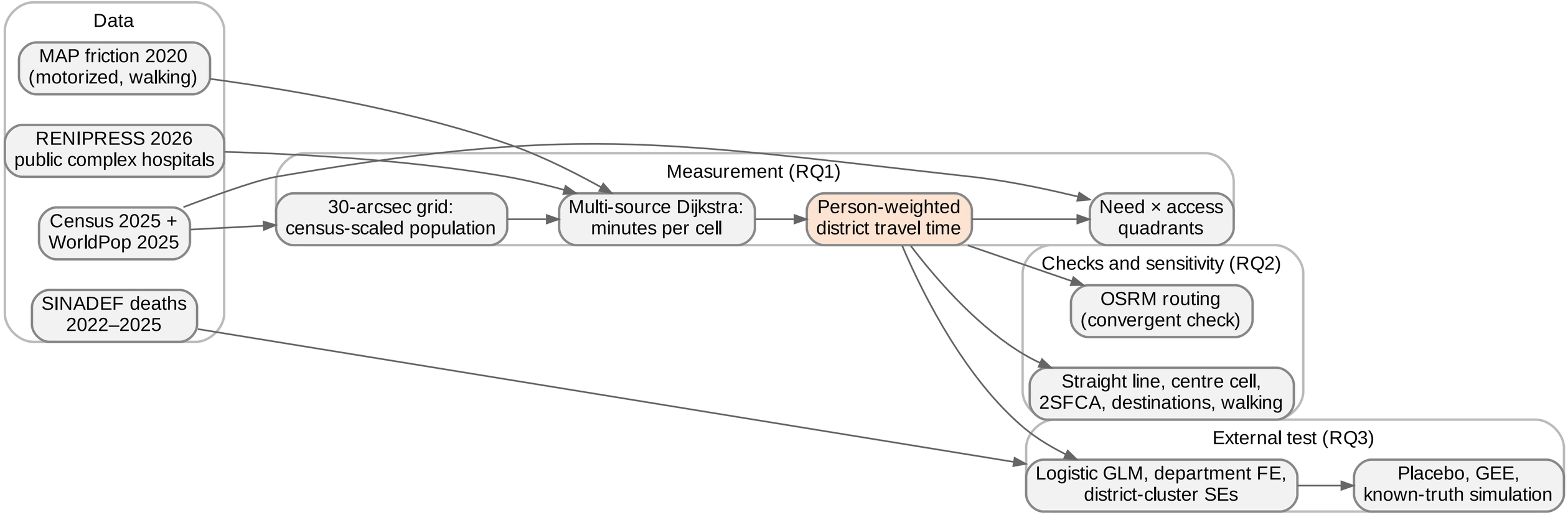
Data
MAP friction 2020 (motorized, walking)
RENIPRESS 2026 public complex hospitals
Census 2025 + WorldPop 2025
SINADEF deaths 2022–2025
Measurement (RQ1)
30-arcsec grid: census-scaled population
Multi-source Dijkstra: minutes per cell
Person-weighted district travel time
Need × access quadrants
Checks and sensitivity (RQ2)
OSRM routing (convergent check)
Straight line, centre cell, 2SFCA, destinations, walking
External test (RQ3)
Logistic GLM, department FE, district-cluster SEs
Placebo, GEE, known-truth simulation